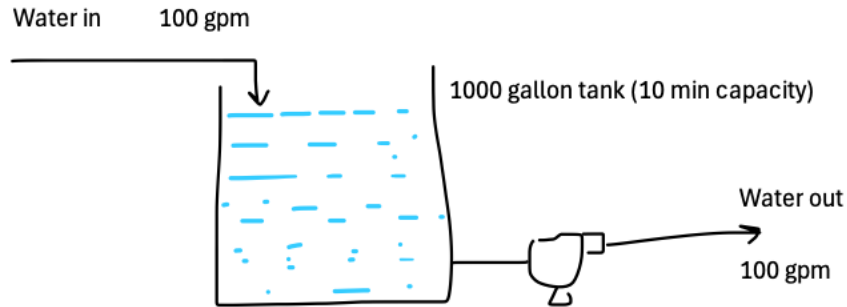


Tank and Pump system

Process Design Criteria – Base Case Report



1. System Description & Process Basis

The system is a clean-water storage and pumping arrangement. Water flows in at 100 gpm into a stainless-steel storage tank and the pump discharges 100 gpm from the tank to downstream process/piping. The tank is designed to hold approximately 10 minutes' storage at 100 gpm flow, i.e., 1,000 gallons nominal at 80% level. The tank is maintained at 80% level by a level transmitter and a feed control valve on the inlet line. The tank is covered (but atmospheric), stainless steel construction, no insulation or heat tracing. All piping is stainless steel. The pump is sized at 10 hp based on the total dynamic head (TDH) for the downstream discharge. The system is installed in Charlotte (Mecklenburg County), North Carolina (USA).

This document sets out the base case numbers, design criteria, material basis, instrumentation, controls, site environmental loads and applicable codes/standards.

2. Major Equipment & Process Data

Item	Description / Specification	Basis / Notes
Inlet flow	100 gpm (nominal)	Clean water feed to tank
Outlet flow	100 gpm (nominal)	Pump discharge rate equals inlet, steady state
Storage tank capacity (nominal)	1,000 gal @ 80% full	$100 \text{ gpm} \times 10 \text{ min} = 1,000 \text{ gal}$; design at 80% full level
Tank percent full for control	80% level maintained	Level transmitter + control feed valve logic
Tank material	Stainless steel	Corrosion-resistant for clean water service
Piping material	Stainless steel	Matches tank, avoids contamination/corrosion
Pump motor size	10 hp	Based on calculated TDH for downstream system (to be validated)
Tank pressure / atmosphere	Covered atmospheric; no special pressurization	Venting, overflow line provided
Tank insulation / heat trace	None	Service is clean ambient water – no freeze/heat control specified

Item	Description / Specification	Basis / Notes
Controls	Level transmitter, feed control valve on inlet, overflow line for high level protection	Ensures control of tank level at 80% and prevents overflow
Nozzles / flanges / manways	Standard flanged nozzles and manway/access as required	Good design practice for maintenance/access
Location	Charlotte, NC, USA	Site conditions for structural & mechanical design

3. Design Criteria / Assumptions

Process and hydraulic design

1. The system is designed for a steady flow of 100 gpm in and 100 gpm out.
2. The tank must provide at least 10 minutes of storage at nominal flow: 1,000 gal.
3. The tank is maintained at 80% full under normal conditions, meaning that the physical capacity must allow this plus some margin.
4. The pump must handle the required flow (100 gpm) and the total dynamic head (TDH) of the downstream system (to be determined in further engineering). Motor size assumed at 10 hp as a base case; actual sizing will depend on hydraulic calculations, efficiency, and safety margins.
5. The tank is atmospheric (no elevated internal pressure) and covered; venting is assumed. The overflow line is sized for safety relief of excessive fill.
6. No insulation or heat tracing is included: ambient conditions must suffice to avoid freezing or overheating; service fluid is clean water. If freezing risk or ambient extremes dictate, add heat trace/insulation in detailed design.
7. All wetted components (tank, piping) are stainless steel, compatible with clean water service.
8. Controls: A level transmitter at the tank, controlling the inlet feed control valve to maintain ~80% level. Manual or automatic overflow protection (overflow line) for high-level condition. Consider low-level alarm and pump interlock if required.
9. Access: Provide manway for inspection and cleaning, sample/vent nozzles, instrumentation ports, drain nozzle.

Materials / mechanical design

- Stainless steel (specify grade, e.g., 304 or 316) for tank shell, supports, piping.
- Corrosion allowances as per clean water service (minimal but follow standard).
- Piping support spacing and design per ASME B31.3 (if considered process piping) or B31.1 as appropriate.
- Welds, flanges, bolts, gaskets compatible with materials and service.
- Tank supports sized for full water load, dead weight + live loads (including site seismic/ wind loads – see section 4).
- Pump and motor located either at base of tank or separate as defined; ensure suction conditions are adequate (NPSH available) and suction piping is sized for 100 gpm.

Instrumentation & Controls

- Level transmitter (e.g., guided-wave radar or ultrasonic) rated for clean water at atmospheric pressure in tank. Calibration around 0-100% level range, with 80% setpoint.
- Feed control valve on inlet water line, pneumatically or electrically actuated, tied to level transmitter signal. Preferably with manual override.
- High-level overflow line (passive) to prevent overfill. Possibly an alarm/high-level switch.
- Consider low-level interlock or alarm (if pump operation depends on minimum level).
- Pump control: On start/stop logic, possibly auto start when outlet demand exists; interlock with tank level (e.g., inhibit pump if tank too low).
- Instrumentation wiring, controls cabinet, local HMI and/or SCADA interface.

Safety / access

- Tank manway, handrails or access stairs if required.
 - Overflow line routed to safe location.
 - Drain nozzle with isolation valve.
 - Vent nozzle sized appropriately.
 - Signage and labeling per site practices.
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4. Site Geographic & Environmental Data – Charlotte, NC

Parameter	Value / Basis	Notes & Source
Location elevation	~ 760 ft (~ 232 m) above sea level	Charlotte elevation ~ 760 ft
Ambient temperature (design)	Min winter ambient ~ 20 °F (–6.7 °C), Max summer ambient ~ 100 °F (37.8 °C)	Use typical Charlotte data; detailed climate data to be referenced.
Relative humidity (typical)	High humidity, especially summer (~70-90 %)	Important for corrosion and equipment ventilation.
Atmospheric pressure	~ 14.7 psia at sea level; at 760 ft maybe ~14.5 psia	For tank vent/pressure design.
Wind load	To be designed per ASCE 7-16 and 2018 NCBC (ex: basic wind speed ~90 mph for Risk Category II) OSFM+1	Structural supports for tank and piping should consider wind uplift/lateral loads.
Snow load	Ground snow load in Charlotte ~ 0 psf (non-snow region) or minimal (e.g., <10 psf) per NCBC Chapter 16 ICC Digital Codes+1	Snow loads are minimal; still check local code.
Seismic load	Seismic design category likely A or B (low seismic hazard) per ASCE 7	Evaluate site-specific; Charlotte region is relatively low seismic.
Flood / flood zone	Verify project site FEMA flood zone; assume not in coastal high hazard zone	Must check actual site.
Design life	Typical mechanical / piping life expectancy ≥ 20-30 years, with corrosion allowance and maintenance.	

Codes & standards applicable for environmental/structural loads:

- 2018 North Carolina Building Code (NCBC) – Chapter 16 Structural Design for loads (snow, wind, seismic). [ICC Digital Codes+1](#)
 - ASCE 7 (Minimum Design Loads for Buildings and Other Structures) for wind, snow, seismic.
 - Local jurisdiction (Charlotte/Mecklenburg) may adopt additional criteria.
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5. Codes, Standards & References

Below is a non-exhaustive list of codes/standards to be used in detailed engineering design of this system.

Discipline	Reference Standard	Remarks
Structural Loads	ASCE 7-16 (or latest) “Minimum Design Loads for Buildings and Other Structures”	Provides wind, snow, seismic loadings.
Building/Structural Code	2018 North Carolina Building Code (NCBC) – especially Chapter 16 “Structural Design”	Governs structural design loads in NC. ICC Digital Codes+1

Discipline	Reference Standard	Remarks
Process Piping	ASME B31.3 “Process Piping”	For stainless steel piping design, supports, stress analysis.
Pressure Vessels / Tanks	ASME Boiler & Pressure Vessel Code (BPVC) Section VIII, Div. 1 (if considered a pressure vessel)	If tank is atmospheric, may not require full pressure-vessel code; verify local jurisdiction.
Storage Tanks	API 650 “Welded Steel Tanks for Oil Storage” (for atmospheric tanks) or other appropriate standard (e.g., AWWA D100 for water tanks)	For design of stainless steel storage tank; adapt to clean water service.
Materials	ASTM A240 (for stainless steel sheet/plate), ASTM A312 (stainless piping)	Material standardization.
Pumps	Hydraulic Institute Standards (e.g., ANSI/HI standards) for pump design, testing	Ensures pump performance and selection.
Electrical / Motor	NFPA 70 (National Electrical Code), IEEE 841 (for motor if severe duty)	For motor and electrical installation.
Instrumentation/Control	ISA standards (e.g., ISA-5.1 Instrumentation Symbols), IEC 61511 (for safety instrumented systems if applicable)	For instrumentation and control system design.
Corrosion / Water Quality	ASTM, AWWA standards for water quality, corrosion control	Ensure compatibility of clean water service.
Tank Supports / Foundation	AISC Steel Construction Manual (if supports steel frames)	Structural design of supports.

6. Process Design Basis – Summary Table

Here is a consolidated table of key design basis parameters for the system:

Parameter	Value	Notes / Comments
Feed flow rate	100 gpm	Nominal steady state.
Discharge flow rate	100 gpm	Matches feed in steady state.
Tank storage time	10 minutes	Basis for storage size.
Tank nominal volume	1,000 gallons	100 gpm × 10 min.
Tank operational fill level	80 %	Controlled level of tank.
Tank material	Stainless steel (specify grade)	Suitable for clean water.
Tank design full volume required	~1,000 gallons ÷ 0.8 = ~1,250 gallons gross theoretical (to allow sump or headspace)	Additional headspace should be included; assume 1,250 gal gross capacity for design.
Tank operating pressure	Atmospheric	Venting to atmosphere; minimal internal pressure.
Tank cover	Yes (covered)	Minimizes contamination and evaporation.
Tank insulation / heat trace	None	Acceptable only if ambient conditions permit; check freezing risk.
Suction and discharge piping material	Stainless steel	Full system compatibility.
Pump motor size	10 hp	Base case; actual sizing to confirm with hydraulic head, pump curves.
Controls – Level	Level transmitter controlling inlet feed valve	Maintain ~80% level.
Controls – Inlet Valve	Automated control valve	Feed water control.
Controls – Overflow protection	Passive overflow line (and high level alarm)	Safety measure.

Parameter	Value	Notes / Comments
Instrumentation – Low level interlock	Recommended (not explicitly specified)	To protect pump suction from low level conditions.
Ambient temperature (design)	–6.7 °C (20 °F) min winter, 37.8 °C (100 °F) max summer	Use for mechanical/piping design.
Wind design speed	e.g., ~90 mph (3-second gust) for Risk Category II per NCBC/ASCE7	Verify with current local adopted map.
Snow load	Minimal (<10 psf) for Charlotte region per NCBC Chapter 16 ICC Digital Codes+1	Confirm local jurisdiction.
Seismic design category	Likely Category A (low)	Verify based on exact site location.
Corrosion allowance	To be determined (e.g., 1/16" for stainless steel; minimal)	Clean water service.
Design life	≥ 20 years	For mechanical/piping equipment.
Access & maintenance	Manway, drain nozzle, vent, sample port	Included in tank design.
Foundation / supports	Sized for dead weight (tank + water), live loads (maintenance), wind/seismic loads	Structural engineer to define.

7. Key Design Notes / Considerations for Further Engineering

- **Tank volume & head-space:** Although the requirement is 1,000 gallons at 80% fill, the gross volume needs a headspace for instrumentation, venting, and possible surge allowance. Consider specifying gross volume as ~1,250 gallons.
- **Pump sizing:** The 10 hp motor is a base assumption; detailed hydraulic calculations (flow, total dynamic head, suction conditions, NPSH, pump efficiency, motor service factor) are required. Also consider variable flow if system demand changes.
- **Suction conditions:** Ensure the tank outlet to pump suction piping is sized to avoid cavitation, maintain required NPSH margin. Provide sufficient tank liquid level above pump suction for gravity feed or consider booster if necessary.
- **Control scheme:** Verify control logic: the level transmitter should continuously monitor tank level; the inlet valve modulates to maintain 80% setpoint. Provide alarms for high and low level, and interlocks to disable pump if tank falls below a safe level.
- **Overflow line:** Size for worst-case feed (100 gpm) plus any transient; route to safe drainage.
- **Materials:** Confirm stainless steel grade (304L, 316L) based on water chemistry, chloride content, etc. Ensure all wetted fasteners, supports, welds compatible.
- **Support/anchorage:** The tank support structure must be designed for the water load (1,000+ gallons = ~8.34 lb/gal × 1,000 = ~8,340 lb + tank weight), plus piping loads, seismic and wind loads as per ASCE 7/NCBC.
- **Environmental exposure:** Although no insulation/heat trace is included, verify freezing risk for Charlotte winter conditions and summer overheating risk. If ambient conditions drop below acceptable, consider heat tracing or insulation.
- **Maintenance & cleaning:** Provide internal access manway, adequate flow for flushing, sample port for water quality, drain business. Consider location of spare parts for pump.
- **Instrumentation & control wiring:** Design cable trays, conduit, control panel location, integration with site BMS/SCADA. Ensure compliance with NFPA 70 for motors and wiring.
- **Foundations:** Provide pad or structural support for tank; consider soil bearing capacity, settlement, seismic base shear, wind uplift anchors.
- **Design margins:** Select pump and piping with margin (e.g., +10-20% flow or head). Provide isolation valves for maintenance.
- **Quality & commissioning:** Specify water quality (clean water, possible treatment?), test procedures for tank (hydrostatic test if required), piping, pump performance test, control loop tuning.
- **Standards compliance:** For example, structural design per ASCE 7-16 and NCBC 2018, piping per ASME B31.3, materials per ASTM, etc. If the local site has additional requirements (e.g., local fire code, environmental permit) ensure compliance.

Capex

1 — Line-item Cost Estimate (detailed)

Design basis: Tank gross ≈ **1,250 gal** (to allow headspace — required 1,000 gal @ 80%); design flow 100 gpm in/out; pump base case 10 hp.

#	Item	Quantity / Units	Unit Cost (USD)	Basis / Comments	Extended Cost (USD)
1	Stainless-steel vertical tank (fabrication & delivery)	1 ea (1,250 gal gross)	\$12/gal (fabrication)	Unit price assumed for small industrial SS tank (includes fabrication, basic fittings). See notes.	15,000
2	Concrete foundation / pad (incl. rebar, formwork)	1 pad (~10'×10'×6")	\$2,500 lump	Small equipment pad sized & installed. Local concrete yard prices used for basis.	2,500
3	Pump + motor + baseplate (centrifugal, 10 hp)	1 set	8,500 lump	Pump with 10-hp motor, baseplate and coupling. Vendor price estimate (industrial midrange).	8,500
4	Suction & discharge piping (stainless steel)	~60 ft (2"–3")	\$20/ft (avg SS)	Material only: stainless pipe (incl. straight lengths). Wall thickness, routing to be confirmed.	1,200
5	Control (modulating) feed control valve	1 ea	7,500 lump	Motorized/modulating control valve, stainless or stainless internals.	7,500
6	Valves & fittings (isolation, check, strainers, flanges)	Assorted	3,000 lump	Ball valves, check valve, flanges/gaskets, strainers, fittings.	3,000
7	Instrumentation & controls	1 set	5,000 lump	Level transmitter (radar/ultrasonic ~\$1k–2k), local switches, junction boxes, basic PLC/relay interface.	5,000
8	Electrical — motor starter, VFD, panel, cables	lump	11,500 lump	VFD (~\$3,500), starter/panel (~\$3,000), cables/conduits (~\$3,000), service tie (~\$2,000). Assumes short run to nearest service.	11,500
9	Earthworks, grading, trenching, backfill	lump	5,000 lump	Excavation for pad and trenches, backfill & compaction.	5,000
10	Site prep, gravel pad, drainage	lump	3,000 lump	Gravel working pad, simple drainage swale, demobilization.	3,000
11	Mechanical & electrical installation labor	lump	25,000 lump	Rigging and set-up of tank, piping installation & welding, pump alignment, instrument wiring, electrical hookup.	25,000
12	Engineering & procurement (PE drawings, specs, procurement support)	15% <i>equipment subtotal</i>	7,005	Applied to key equipment subtotal (typical engineering fee for basic design).	7,005
13	Commissioning & testing	lump	5,000 lump	Start-up, loop checks, pump curve run, control tuning.	5,000
14	Spares & consumables	lump	2,000 lump	Basic spare seals, gaskets, instrumentation spare parts.	2,000
	Subtotal (pre-contingency)				101,205
	Contingency (30%)			To cover unknowns, mobilization, site conditions, market variability.	30,362

#	Item	Quantity / Units	Unit Cost (USD)	Basis / Comments	Extended Cost (USD)
	Total Estimate (installed)			All items included above	131,567 ≈ \$132k

(All values rounded for presentation.)

2 — Key assumptions & rationale

- **Tank sizing:** You specified 1,000 gal at 80% level. To achieve 1,000 gal at 80% filled, gross volume ≈ 1,250 gal. I used 1,250 gal for fabrication pricing. Tank geometry assumed small vertical cylindrical tank (roughly 6 ft dia × 6 ft high). If a different geometry or larger manway is required, cost will change.
- **Tank unit cost:** Industry vendor/market data for small stainless storage tanks vary widely. I used **\$~12/gal** (mid-range for small fabricated SS tanks with typical fittings). Sources show small tanks can be inexpensive for low spec, but industrial tanks vary; budget range for stainless tanks can be several dollars to tens of dollars per gallon depending on grade and complexity.
- **Pump:** 10-hp centrifugal pump with stainless wet end or corrosion-resistant materials and 10-hp TEFC motor. Price includes vendor standard pump, motor, baseplate and coupling. Premium materials, special seals (mechanical seals) or certified performance testing will increase cost.
- **Piping:** Estimated 60 ft of stainless piping (2"–3") as a short campus/plant run. Unit cost used \$20/ft (a mid-range figure between ~\$8–\$30/ft depending on grade & schedule). Fittings and labor for welds included separately in valves/fittings and installation labor.
- **Control valve:** Modulating control valve (stainless internals) assumed — these can be significant cost drivers (ranges \$3k–\$15k based on type, size and actuators). I used \$7,500 as a reasonable mid-range.
- **Electrical:** Includes VFD (recommended for pump speed control), starter/panelwork, short cable run and junctioning. If utility run distance is long, transformer upgrades or utility fees will increase cost significantly.
- **Civil / earthworks:** Greenfield site; no prior infrastructure. Estimate assumes normal soil and minimal rock. If rock removal, deeper foundation or permitting is required, costs increase.
- **Installation & labor:** Mechanical & electrical installation lumped based on small system install typical hours (rigging, setting, piping, welding, alignment, instrument hookup).
- **Engineering & commissioning:** Includes PE-stamped drawings at a basic level, procurement assistance and site commissioning.
- **Contingency:** 30% applied to subtotal — appropriate for conceptual estimates on greenfield projects where many items remain to be detailed.
- **Exclusions:** land purchase, long utility extension (beyond short tie), permits/fees, detailed site remediation, instrumentation advanced integration to a plant SCADA (beyond basic I/O), taxes & bonds, environmental permitting, long lead customization, specialized crane mobilization beyond standard rigging. If any of those apply, add separately.

3 — Unit-cost references & supporting sources

(Used to form the mid-range assumptions above — these are *illustrative* references.)

- Stainless-steel tank pricing examples and industry pages (typical small tank price ranges). Example: Tank Depot / vendor pages showing wide range of stainless tank prices. [Tank Depot](#)
- Pump price lists and industrial pump vendor catalogs (price guides for centrifugal pumps). Example Price Pump catalog. [pricepump.com](#)
- Stainless steel pipe price guidance: typical 316 SS pipe costs range roughly **\$8–\$30/ft** depending on size & schedule; used \$20/ft as midrange. [EAGLE+1](#)

- Concrete cost (Charlotte, NC) — recent local ranges **\$129–\$155 per cubic yard** for delivered concrete; pad installed cost used in lumped pad price. [HomeBlue+1](#)
- Level instrumentation price guidance: ultrasonic sensors ~\$500–\$2,000; radar trending similar. Used \$2,000 for a robust unit. [Emerson Automation Experts](#)
- Excavation / earthwork general unit guidance (wide range depending on soil): Homewyse / HomeGuide estimates used for basis. [Homewyse+1](#)

Note: these sources indicate wide market variability — use them to justify the mid-range numbers above. For procurement budgeting, obtain 3 vendor quotes.

4 — Cost sensitivity & next steps (how to firm the budget)

The largest single sensitivities for this system are:

1. **Tank specification** — grade of stainless (304 vs 316), manway size, internal baffles, special finishes — can double or more the tank cost.
2. **Control valve and instrumentation** — modulating valve + actuator + positioner is often a major cost item; seller quotes vary widely.
3. **Electrical / utility service** — if a three-phase service is not nearby and utility extension is required (hundreds of feet or transformer), costs can jump substantially (utility extensions are commonly \$7–\$10/ft or more to tie to nearest line). [ProfitableVenture](#)
4. **Site conditions** — rock excavation, erosion controls, stormwater management, permitting and site grading will materially affect civil scope and cost.
5. **Materials market volatility** — stainless steel and motor/pump pricing can vary with market.

Recommended next steps to firm estimate (in priority order):

1. Obtain **actual vendor quotes** for: stainless tank (specify grade), pump (with curve and materials), control valve (size and materials), stainless piping & fittings.
 2. Conduct a **site visit / geotechnical** to confirm soil, depth to rock, flooding, and to accurately size pad/foundation.
 3. Confirm **utility availability** (distance to 3-phase supply, capacity and any transformer/utility fees). If utility run >100 ft, budget separately.
 4. Produce a simple **P&ID and layout** so piping lengths, insulation/valving and instrumentation counts are exact.
 5. If required, add items excluded above (permits, erosion control, paved access, fencing, lighting).
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5 — Quick alternative budget scenarios (for planning)

- **Conservative (high-side) budget:** allow **\$160k – \$200k** if: 316L tank, long utility run (transformer), significant rock excavation, higher-end modulating valve & specialty pump materials, heavier foundation.
- **Low-cost (if competitive procurement and favorable site):** **\$80k – \$100k** if: tank is modest (304), short piping runs, existing utility nearby, uncomplicated pad and minimal civil work.